

ENERGY ECONOMICS

Introduction

- In the process of energy management, at some stage, **investment would be required for reducing the energy consumption of a process or utility.**
- Investment would be **required for modifications/ retrofitting and for incorporating new technology.**
- It is essential to identify the benefits of the proposed measure with reference to **not only energy savings** but also other associated benefits such as **increased productivity, improved product quality etc.**

The cost involved in the proposed measure should be captured in totality viz.

- Direct project cost
- Additional operations and maintenance cost
- Training of personnel on new technology etc.

Based on the above, the investment analysis can be carried out by the techniques explained in the later slides

Investment Need

To persuade your organization to commit itself to a program of investment in energy efficiency, you need to demonstrate:

- The size of the energy problem it currently faces
- The technical and good housekeeping measure available to reduce waste
- The predicted return on any investment
- The real returns achieved on particular measures over time.

The need for investments in energy conservation can arise under following circumstances

- For new equipment, process improvements etc.
- To provide staff training
- To implement or upgrade the energy information system

Criteria

Any investment has to be seen as an addition and not as a substitute for having effective management practices for controlling energy consumption throughout your organization.

Spending money on technical improvements for energy management cannot compensate for inadequate attention to gaining control over energy consumption. Therefore, before you make any investments, it is important to ensure that

- You are getting the best performance from existing plant and equipment
- Your energy charges are set at the lowest possible tariffs
- You are consuming the best energy forms - fuels or electricity - as efficiently as possible
- Good housekeeping practices are being regularly practiced.

When listing investment opportunities, the following criteria need to be considered:

- The energy consumption per unit of production of a plant or process
- The current state of repair and energy efficiency of the building design, plant and services, including controls
- The quality of the indoor environment - not just room temperatures but indoor air quality and air change rates, drafts, under and overheating including glare, etc.
- The effect of any proposed measure on staff attitudes and behaviour.

Financial Analysis

In most respects, investment in energy efficiency is no different from any other area of financial management. So when your organization first decides to invest in increasing its energy efficiency it should apply exactly the same criteria to reducing its energy consumption as it applies to all its other investments. It should not require a faster or slower rate of return on investment in energy efficiency than it demands elsewhere.

The basic criteria for financial investment appraisal include:

1. **Simple Payback** - a measure of how long it will be before the investment makes money, and how long the financing term needs to be
2. **Return on Investment (ROI) and Internal Rate of Return (IRR)** - measure that allow comparison with other investment options
3. **Net Present Value (NPV) and Cash Flow** - measures that allow financial planning of the project and provide the company with all the information needed to incorporate energy efficiency projects into the corporate financial system.

Simple Pay Back Period:

Simple Payback Period (SPP) represents, as a first approximation; the time (number of years) required to recover the initial investment (First Cost), considering only the Net Annual Saving:

The simple payback period is usually calculated as follows:

$$\text{Simple payback period} = \frac{\text{First cost}}{\text{Yearly benefits} - \text{Yearly costs}}$$

Example

Simple payback period for a continuous Deodorizer that costs Rs.60 lakhs to purchase and install, Rs.1.5 lakhs per year on an average to operate and maintain and is expected to save Rs.20 lakhs by reducing steam consumption (as compared to batch deodorizers), may be calculated as follows:

According to the payback criterion, the shorter the payback period, the more desirable the project

$$\begin{aligned}\text{Simple payback period} &= \frac{\text{First cost}}{\text{Yearly benefits} - \text{Yearly costs}} \\ &= \frac{60}{20 - 1.5} = 3 \text{ years } 3 \text{ months}\end{aligned}$$

Advantages

A widely used investment criterion, the payback period seems to offer the following advantages:

- It is simple, both in concept and application. Obviously a shorter payback generally indicates a more attractive investment. It does not use tedious calculations.
- It favours projects, which generate substantial cash inflows in earlier years, and discriminates against projects, which bring substantial cash inflows in later years but not in earlier years.

Limitations

- It fails to consider the time value of money. Cash inflows, in the payback calculation, are simply added without suitable discounting. This violates the most basic principle of financial analysis, which stipulates that cash flows occurring at different points of time can be added or subtracted only after suitable compounding/discounting.
- It ignores cash flows beyond the payback period. This leads to discrimination against projects that generate substantial cash inflows in later years.

To illustrate, consider the cash flows of two projects, A and B:

The payback criterion prefers A, which has a payback period of 3 years, in comparison to B, which has a payback period of 4 years, even though B has very substantial cash inflows in years

Investment	Rs. (100,000)	Rs.(100,000)
Savings in Year	Cash Flow of A	Cash flow of B
1	50,000	20,000
2	30,000	20,000
3	20,000	20,000
4	10,000	40,000
5	10,000	50,000
6	-	60,000

- It is a measure of a project's capital recovery, not profitability.
- Despite its limitations, the simple payback period has advantages in that it may be useful for evaluating an investment.

Time Value of Money

- A project usually entails an investment for the **initial cost of installation, called the capital cost**, and a series of **annual costs** and/or cost savings (i.e. operating, energy, maintenance, etc.) throughout the life of the project.
- To assess project feasibility, all these present and future cash flows must be equated to a common basis.
- **The problem with equating cash flows which occur at different times is that the value of money changes with time.** The method by which these various cash flows are related is called **discounting**, or the **present value concept**.

The relationship between present and future value is determined as follows:

$$\text{Future Value (FV)} = \text{NPV} (1 + i)^n \quad \text{or} \quad \text{NPV} = \text{FV} / (1+i)^n$$

Where

FV = Future value of the cash flow

NPV = Net Present Value of the cash flow

i = Interest or discount rate

n = Number of years in the future

Return on Investment (ROI)

ROI expresses the "annual return" from the project as a percentage of capital cost.

The annual return takes into account **the cash flows over the project life and the discount rate** by converting the total present value of ongoing cash flows to an equivalent annual amount over the life of the project, which can then be compared to the capital cost.

This is a broad indicator of the annual return expected from initial capital investment, expressed as a percentage:

$$\text{ROI} = \frac{\text{Annual Net Cash Flow}}{\text{Capital Cost}} \times 100$$

ROI must always be higher than cost of money (interest rate); the greater the return on investment better is the investment.

Limitations

- It does not take into account the time value of money.
- It does not account for the variable nature of annual net cash inflows.

Net Present Value

The net present value (NPV) of a project is equal to the sum of the present values of all the cash flows associated with it. Symbolically,

$$\text{NPV} = \frac{\text{CF}_0}{(1 + \kappa)^0} + \frac{\text{CF}_1}{(1 + \kappa)^1} + \dots + \frac{\text{CF}_n}{(1 + \kappa)^n} = \sum_{t=0}^n \frac{\text{CF}_t}{(1 + \kappa)^t}$$

Where NPV = Net Present Value

CF_t = Cash flow occurring at the end of year 't' (t=0,1,...n)

n = life of the project

k = Discount rate

The discount rate (k) employed for evaluating the present value of the expected future cash flows should reflect the risk of the project.

Example

To illustrate the calculation of net present value, consider a project, which has the following cash flow stream:

Investment	Rs. (1,000,000)
Saving in Year	Cash flow
1	200,000
2	200,000
3	300,000
4	300,000
5	350,000

The cost of capital, κ , for the firm is 10 per cent. The net present value of the proposal is:

$$\begin{aligned} \text{NPV} = & - \frac{1,000,000}{(1.10)^0} + \frac{200,000}{(1.10)^1} + \frac{200,000}{(1.10)^2} \\ & + \frac{300,000}{(1.10)^3} + \frac{300,000}{(1.10)^4} + \frac{350,000}{(1.10)^5} = (5,273) \end{aligned}$$

The decision rule associated with the net present value criterion is: "**accept the project If the net present value is positive and reject the project if the net present value is negative**".

Advantages

The net present value criterion has considerable merits.

- It takes into account the time value of money.
- It considers the cash flow stream in its project life.

Internal Rate of Return

This method calculates the rate of return that the investment is expected to yield. The internal rate of return (IRR) method expresses each investment alternative in terms of a rate of return (a compound interest rate).

The expected rate of return is the interest rate for which total discounted benefits become just equal to total discounted costs (i.e net present benefits or net annual benefits are equal to zero, or for which the benefit / cost ratio equals one). The criterion for selection among alternatives is to choose the investment with the highest rate of return.

The rate of return is usually calculated by a process of trial and error, whereby the net cash flow is computed for various discount rates until its value is reduced to zero.

The internal rate of return (IRR) of a project is the discount rate, which makes its net present value (NPV) equal to zero. It is the discount rate in the equation:

$$0 = \frac{CF_0}{(1 + \kappa)^0} + \frac{CF_1}{(1 + \kappa)^1} + \dots + \frac{CF_n}{(1 + \kappa)^n} = \sum_{t=0}^n \frac{CF_t}{(1 + \kappa)^t}$$

where CF_t = cash flow at the end of year "t"

κ = discount rate

n = life of the project.

CF_t value will be negative if it is expenditure and positive if it is savings.

In the net present value calculation we assume that the discount rate (cost of capital) is known and determine the net present value of the project. In the internal rate of return calculation, we set the net present value equal to zero and determine the discount rate (internal rate of return), which satisfies this condition.

To illustrate the calculation of internal rate of return, consider the cash flows of a project:

Year	0	1	2	3	4
Cash flow	(100,000)	30,000	30,000	40,000	45,000

The internal rate of return is the value of " κ " which satisfies the following equation:

$$100,000 = \frac{30,000}{(1 + \kappa)^1} + \frac{30,000}{(1 + \kappa)^2} + \frac{40,000}{(1 + \kappa)^3} + \frac{45,000}{(1 + \kappa)^4}$$

The calculation of "k" involves a process of trial and error. We try different values of "k" till we find that the right-hand side of the above equation is equal to 100,000.

Let us, to begin with, try $k = 15$ per cent. This makes the right-hand side equal to:

$$\frac{30,000}{(1.15)} + \frac{30,000}{(1.15)^2} + \frac{40,000}{(1.15)^3} + \frac{45,000}{(1.15)^4} = 100,802$$

This value is slightly higher than our target value, 100,000. So we increase the value of k from 15 per cent to 16 per cent. (In general, a higher k lowers and a smaller k increases the righthand side value). The right-hand side becomes:

$$\frac{30,000}{(1.16)} + \frac{30,000}{(1.16)^2} + \frac{40,000}{(1.16)^3} + \frac{45,000}{(1.16)^4} = 98,641$$

Since this value is now less than 100,000, we conclude that the value of k lies between 15 per cent and 16 per cent. For most of the purposes this indication suffices.

Advantages

A popular discounted cash flow method, the internal rate of return criterion has several advantages:

- It takes into account the time value of money.
- It considers the cash flow stream in its entirety.
- It makes sense to businessmen who prefer to think in terms of rate of return and find an absolute quantity, like net present value, somewhat difficult to work with.

Limitations

- The internal rate of return figure cannot distinguish between lending and borrowing and hence a high internal rate of return need not necessarily be a desirable feature.

Example

Calculate the internal rate of return for an economizer that will cost Rs.500,000, will last 10 years, and will result in fuel savings of Rs.150,000 each year.

Find the i that will equate the following:

$$\text{Rs.}500,000 = 150,000 \times \text{PV} (A = 10 \text{ years}, i = ?)$$

To do this, calculate the net present value (NPV) for various i values, selected by visual inspection;

$$\begin{aligned} \text{NPV } 25\% &= \text{Rs.}150,000 \times 3.571 - \text{Rs.}500,000 \\ &= \text{Rs.}35,650 \end{aligned}$$

$$\begin{aligned}\text{NPV } 30\% &= \text{Rs.}150,000 \times 3.092 - \text{Rs. } 500,000 \\ &= -\text{Rs. } 36,200\end{aligned}$$

For $i = 25$ per cent, net present value is positive; $i = 30$ per cent, net present value is negative.

Thus, for some discount rate between 25 and 30 per cent, present value benefits are equated to present value costs. To find the rate more exactly, one can interpolate between the two rates as follows:

$$\begin{aligned}i &= 0.25 + (0.30-0.25) \times 35650 / (35650 + 36200) \\ &= 0.275, \text{ or } 27.5 \text{ percent}\end{aligned}$$